Command Execution and Purpose

S. No.	Linux Command	Purpose	Example Command	Expected Result
1	`getfacl`	Displays detailed file access control lists (ACLs).	`getfacl /etc/hostname`	Shows file ownership and ACL permissions.
2	`setfacl`	Grants specific users or groups additional file permissions.	`sudo setfacl -m u:nobody:r /tmp/linux_acl.txt`	Grants `nobody` read access to the test file.
3	`namei`	Shows permissions and ownership along a complete path.	`namei -l /etc/hostname`	Displays each directory and file component with permissions.
4	`getcap`	Lists Linux capabilities assigned to executables.	`getcap /usr/bin/ping`	Displays any capabilities assigned to `ping`.
5	`lscap`	Lists capabilities supported by the running kernel.	`lscap`	Displays supported Linux capability names, if the utility is installed.
6	`chattr`	Sets special filesystem attributes,	`sudo chattr +i /tmp/linux_attr.txt`	Marks the test file immutable on

		such as the immutable flag.		supported filesystems.
7	`lsattr`	Displays special filesystem attributes.	`lsattr /tmp/linux_attr.txt`	Shows the file's attributes, including the immutable flag.
8	`fallocate`	Allocates disk space for a file without writing all its contents.	`fallocate -l 1M /tmp/linux_space.bin`	Creates a file with a logical size of 1 MiB.
9	`ionice`	Displays or adjusts a process's I/O scheduling priority.	`ionice -c 3 -p $$`	Sets the current shell's I/O priority to idle where supported.
10	`prlimit`	Displays or modifies process resource limits.	`prlimit --pid $$`	Shows resource limits for the current shell process.

LINUX COMMANDS

```
abishek1703@AAAD:~$ getfacl /etc/hostname
getfacl: Removing leading '/' from absolute path names
# file: etc/hostname
# owner: root
# group: root
user::rw-
group::r--
other::r--

abishek1703@AAAD:~$ printf 'ACL test data\n' > /tmp/linux_acl.txt
abishek1703@AAAD:~$ chmod 600 /tmp/linux_acl.txt
abishek1703@AAAD:~$ sudo setfacl -m u:nobody:r /tmp/linux_acl.txt
abishek1703@AAAD:~$ getfacl /tmp/linux_acl.txt
getfacl: Removing leading '/' from absolute path names
# file: tmp/linux_acl.txt
# owner: abishek1703
# group: abishek1703
user::rw-
user:nobody:r--
group::---
mask::r--
other::---

abishek1703@AAAD:~$ sudo -u nobody cat /tmp/linux_acl.txt
ACL test data
abishek1703@AAAD:~$ namei -l /etc/hostname
f: /etc/hostname
drwxr-xr-x root root /
drwxr-xr-x root root etc
-rw-r--r-- root root hostname
abishek1703@AAAD:~$ getcap /usr/bin/ping
/usr/bin/ping cap_net_raw=ep
abishek1703@AAAD:~$ getconf OPEN_MAX
10240
abishek1703@AAAD:~$ touch /tmp/linux_attr.txt
abishek1703@AAAD:~$ sudo chattr +i /tmp/linux_attr.txt
abishek1703@AAAD:~$ lsattr /tmp/linux_attr.txt
----i---------------- /tmp/linux_attr.txt
abishek1703@AAAD:~$ sudo chattr -i /tmp/linux_attr.txt
abishek1703@AAAD:~$
```

```
abishek1703@AAAD:~$ lsattr /tmp/linux_attr.txt
---------------------- /tmp/linux_attr.txt
abishek1703@AAAD:~$ fallocate -l 1M /tmp/linux_space.bin
abishek1703@AAAD:~$ ls -lh /tmp/linux_space.bin
-rw-r--r-- 1 abishek1703 abishek1703 1.0M Oct 29 15:51 /tmp/linux_space.bin
abishek1703@AAAD:~$ ionice -c 3 -p $$
abishek1703@AAAD:~$ prlimit --pid $$
RESOURCE   DESCRIPTION                             SOFT      HARD UNITS
AS         address space limit                unlimited unlimited bytes
CORE       max core file size                         0 unlimited bytes
CPU        CPU time                           unlimited unlimited seconds
DATA       max data size                      unlimited unlimited bytes
FSIZE      max file size                      unlimited unlimited bytes
LOCKS      max number of file locks held      unlimited unlimited locks
MEMLOCK    max locked-in-memory address space  67108864  67108864 bytes
MSGQUEUE   max bytes in POSIX mqueues            819200    819200 bytes
NICE       max nice prio allowed to raise             0         0
NOFILE     max number of open files               10240   1048576 files
NPROC      max number of processes                31189     31189 processes
RSS        max resident set size              unlimited unlimited bytes
RTPRIO     max real-time priority                     0         0
RTTIME     timeout for real-time tasks        unlimited unlimited microsecs
SIGPENDING max number of pending signals          31189     31189 signals
STACK      max stack size                       8388608 unlimited bytes
abishek1703@AAAD:~$
```